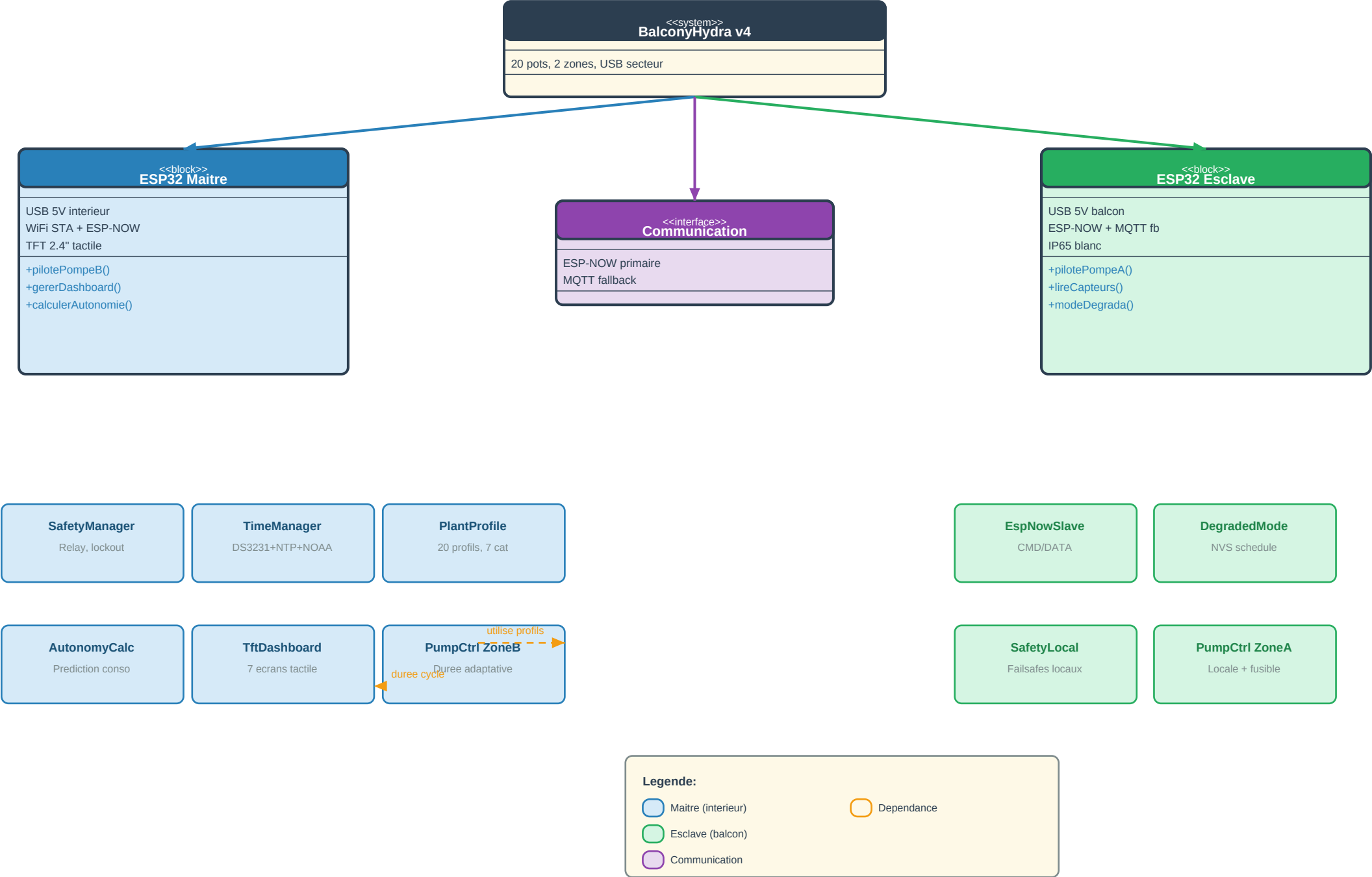
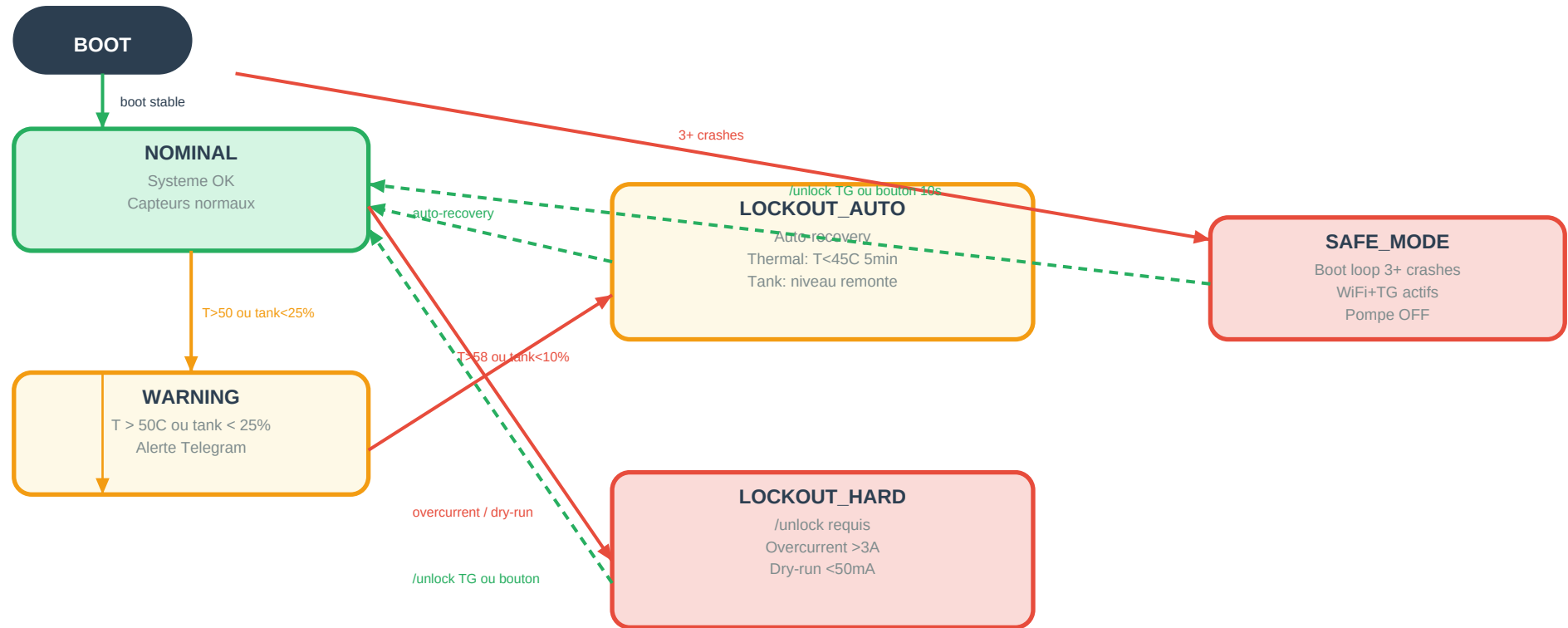


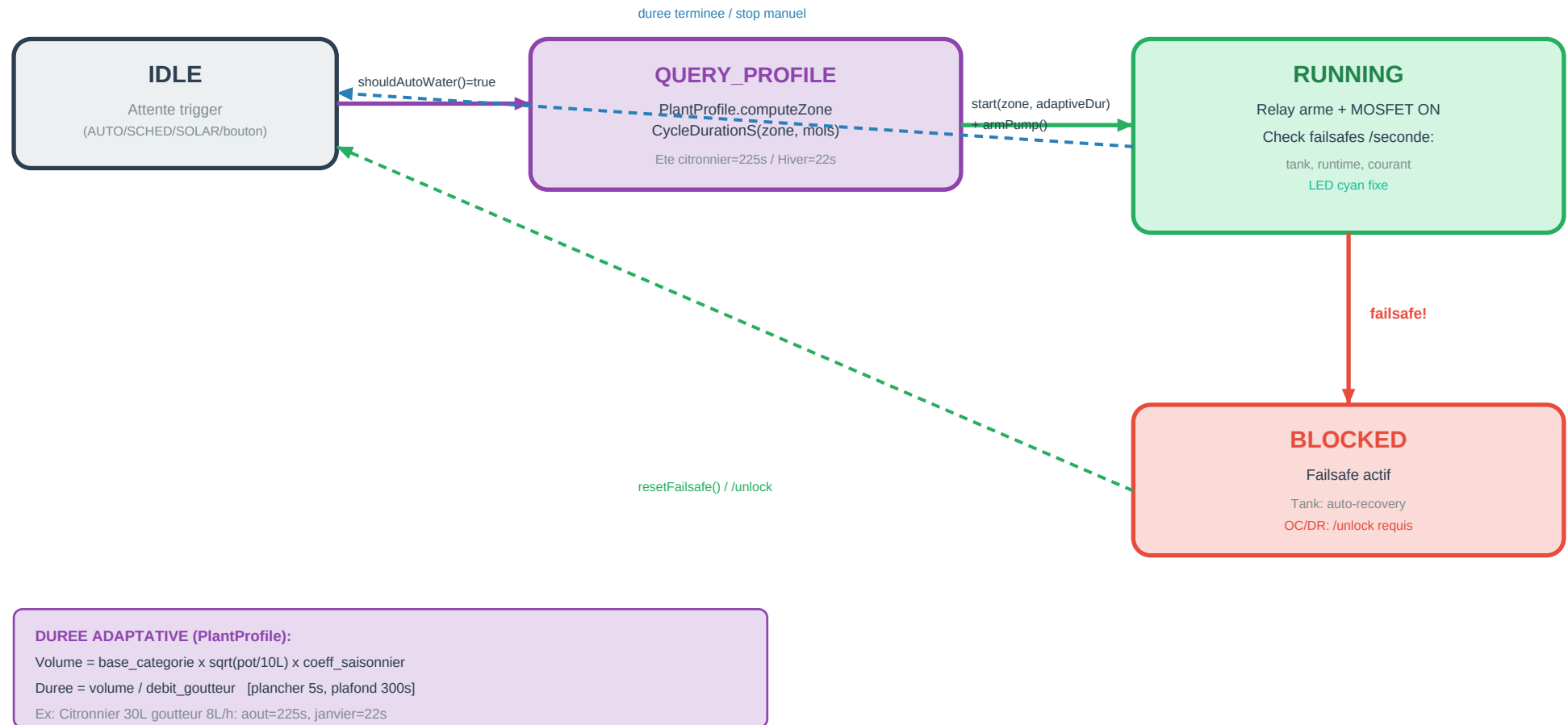


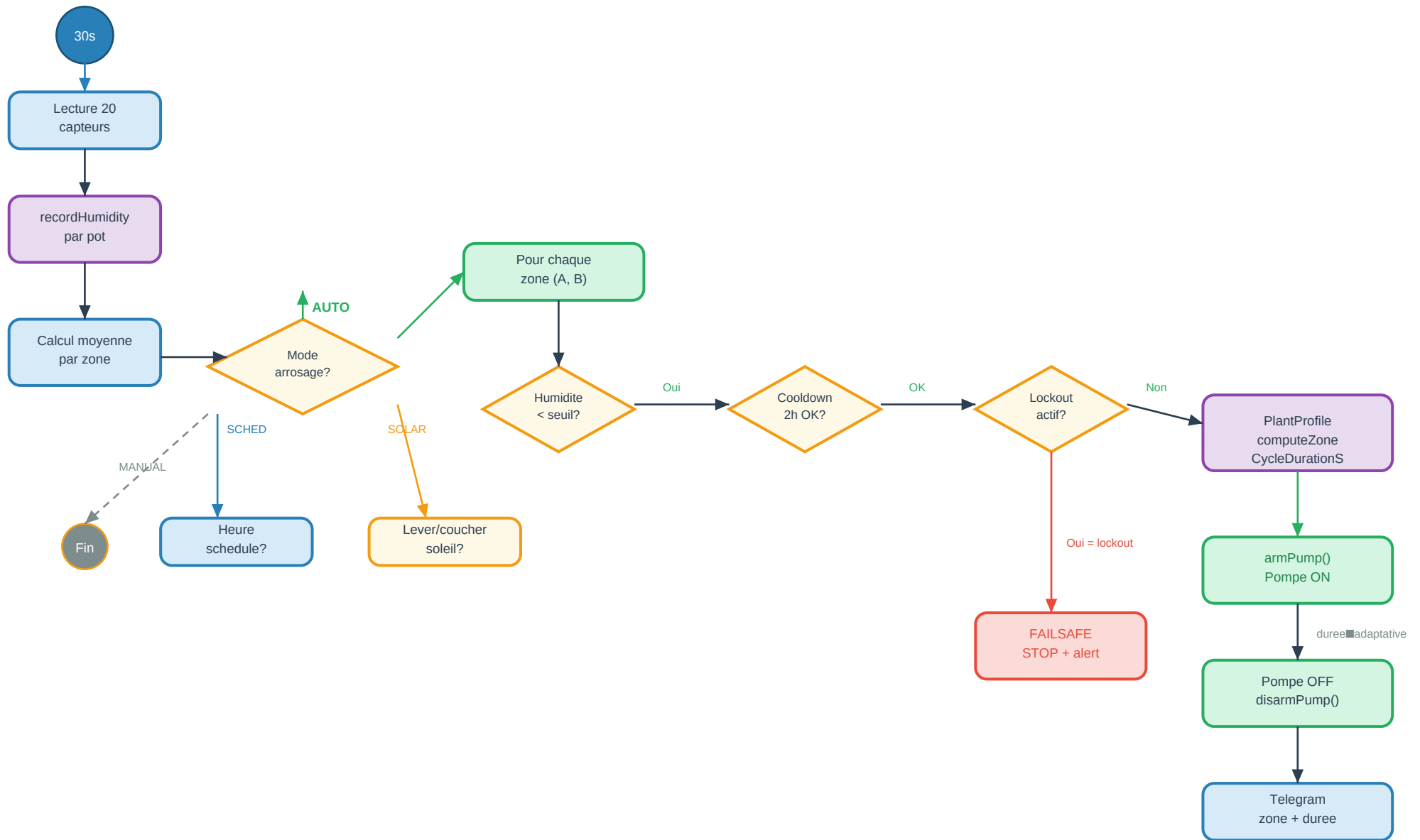
**AUTO-RECOVERY (se rearme seul):**

- Thermal: T redescend <45C stable 5min
- Tank: niveau remonte >10%
- WiFi: reconnexion backoff exponentiel

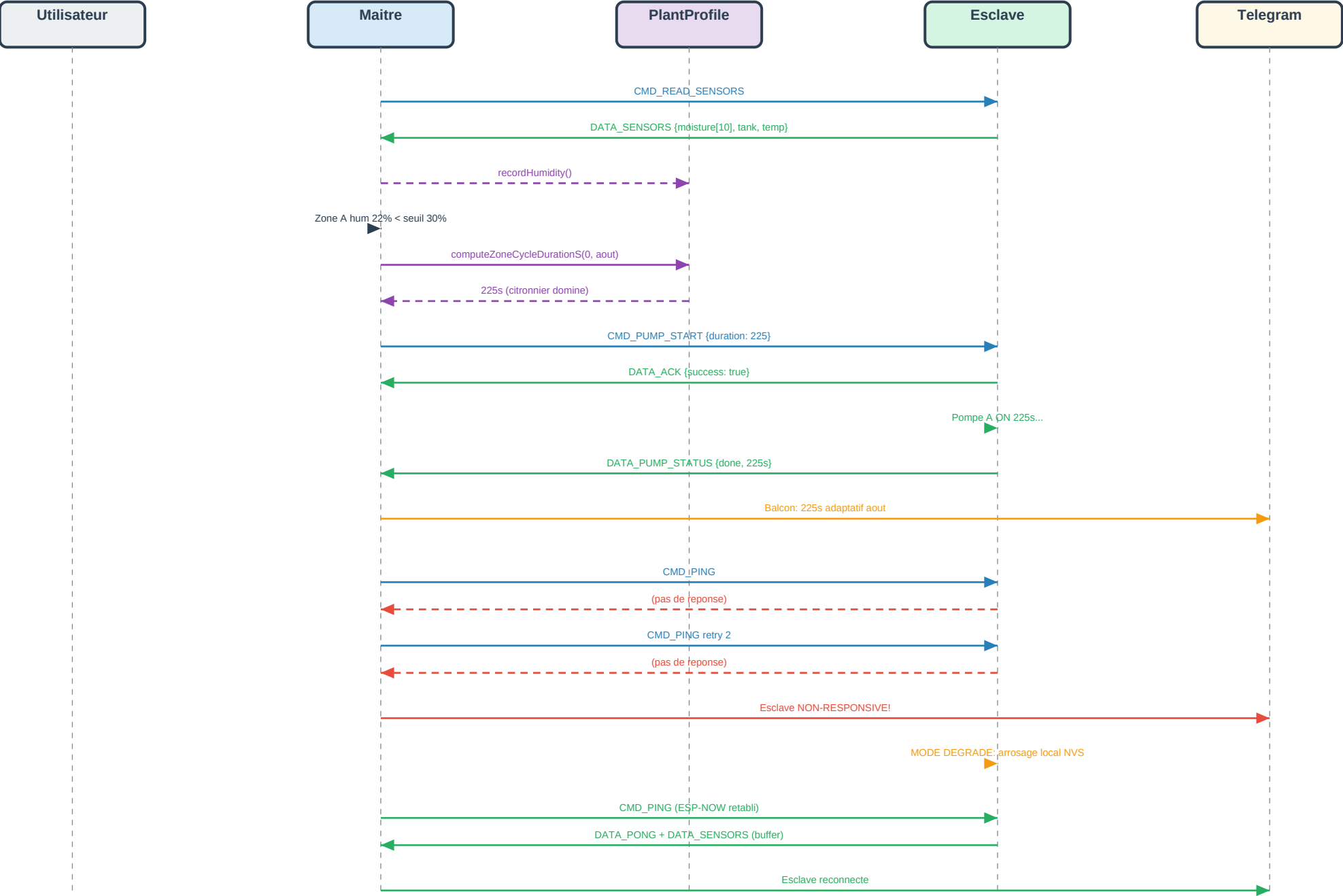
HARD LOCKOUT (/unlock requis):

- Surintensitee pompe >3A
- Marche a sec pompe <50mA
- Boot crash 3+ → safe mode





Cycle normal, arrosage adaptatif, failsafe, perte de comm, recovery



<<system>> Balcony Hydra v4



Utilisateur

Configuration

Configurer WiFi (TFT)

Créer profil hydrique

Choisir mode arrosage

Prediction

Calculer autonomie N jours

Estimer conso/zone

Detecter deficit stockage

Apprendre assechement

Arrosage

Arrosage AUTO adaptatif

Arrosage programme

Arrosage solaire

Arrosage manuel

Securite

Unlock distant /unlock

Etat securite /safety

Reset failsafe

Monitoring

Dashboard TFT 7 ecrans

Alertes Telegram

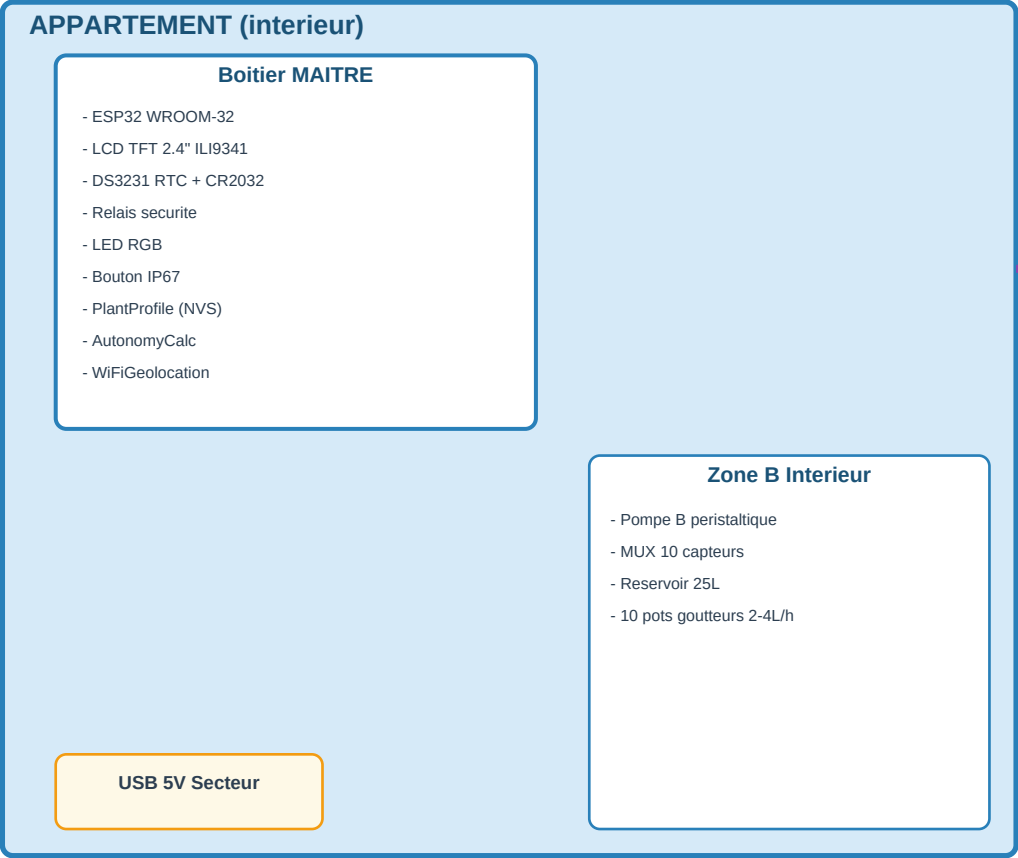
MQTT cloud



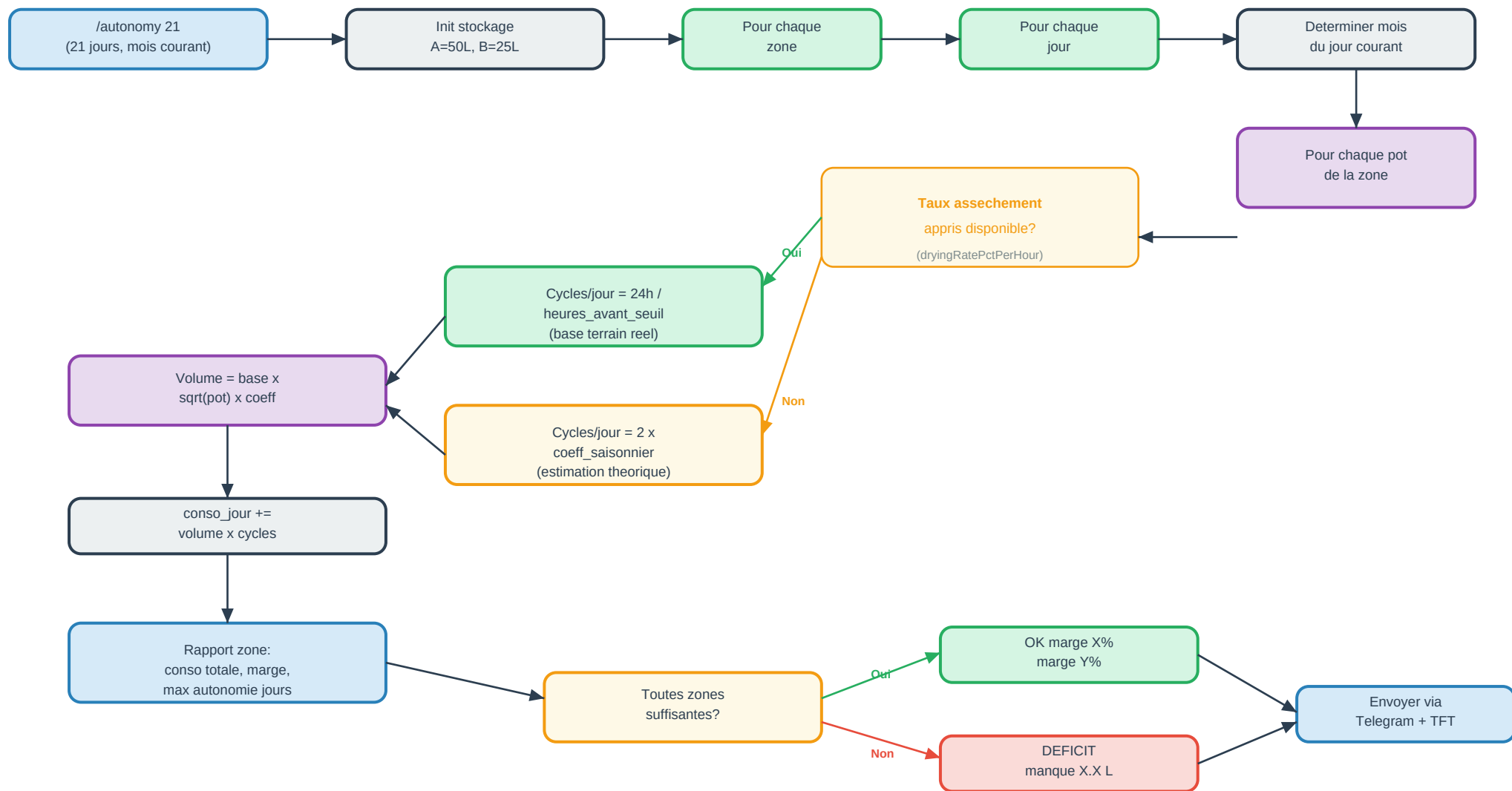
Plantes



Horloge/Soleil



ESP-NOW
+MQTT fb



Esclave perd le maître, continue d'arroser seul, puis recovery

